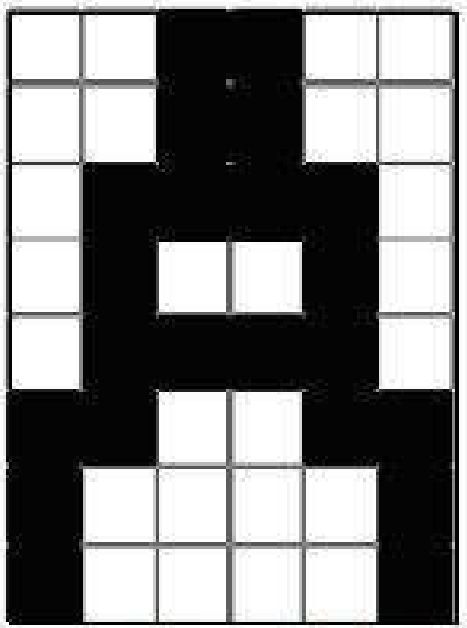
Understanding Image as Data

DIGITAL IMAGES

Images are treated as data by digitally converting them into a grid of pixels, where each pixel has a numerical value representing its colour and intensity

A digital image is a representation of visual data as a *d*-dimensional array of numerical values



Digitised "*A*"

0	0	1	1	0	0
0	0	1	1	0	0
0	1	1	1	1	0
0	1	0	0	1	0
0	1	1	1	1	0
1	1	0	0	1	1
1	0	0	0	0	1
1	0	0	0	0	1

Matrix form of "*A*"

Here, the intensity for each pixel is either 0 or 1 (pixel is either "on" or "off")

However, the intensity range can vary (e.g., 0-255)

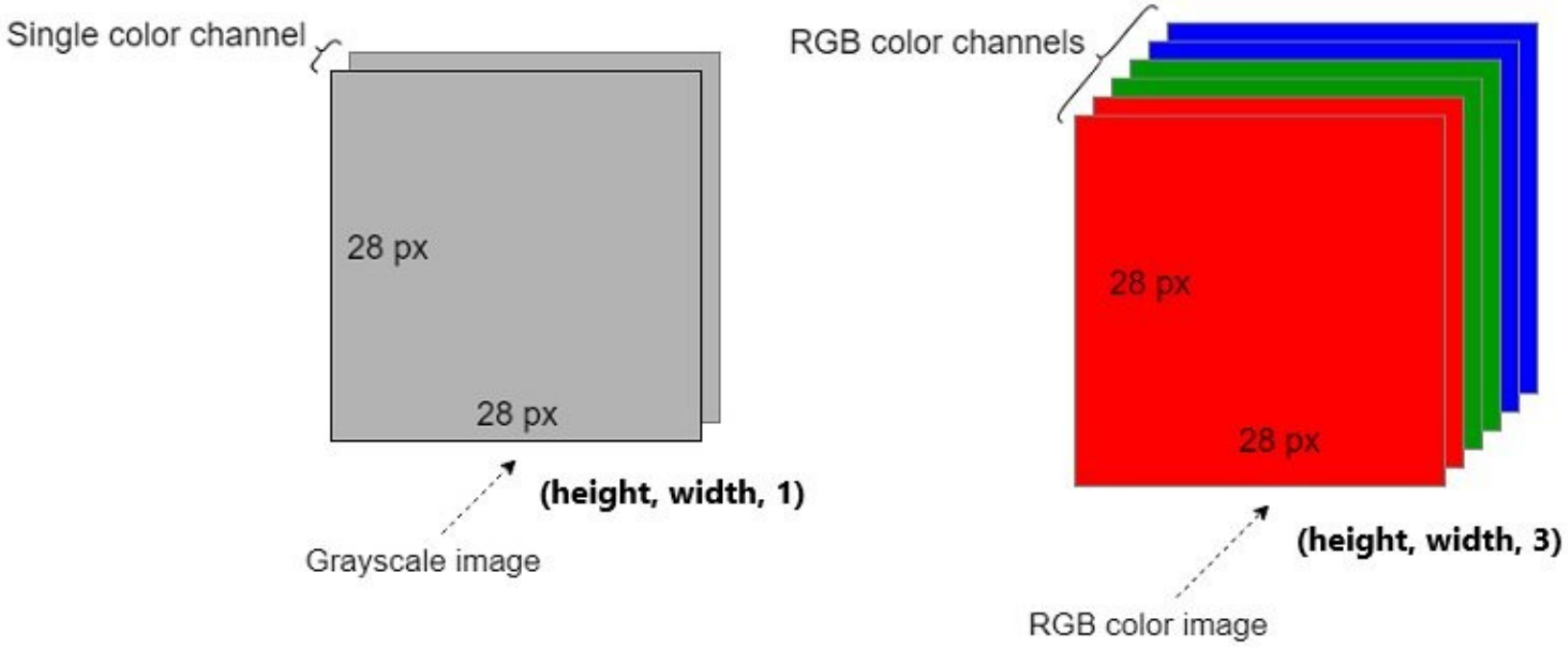
COLOUR CHANNELS

- An image can be grayscale (single channel) or may contain colours (multiple channels)
- The **RGB colour model** is the most common colour model used in digital image processing, consisting of 3 channels: Red, Green and Blue
- Different combinations of these colours in varying intensities create other colours
- Some other common colour models are
  - **CMYK** (Cyan Magenta Yellow Black) Mapping, commonly used in printers
  - **HSV** (Hue Saturation Value), which uses colours in the way our eyes perceive them



Each pixel in the array is now denoted as a set of (R,G,B) intensity values (0 to 255), each value denoting the intensity for one of the three colours

IMAGE DIMENSIONS

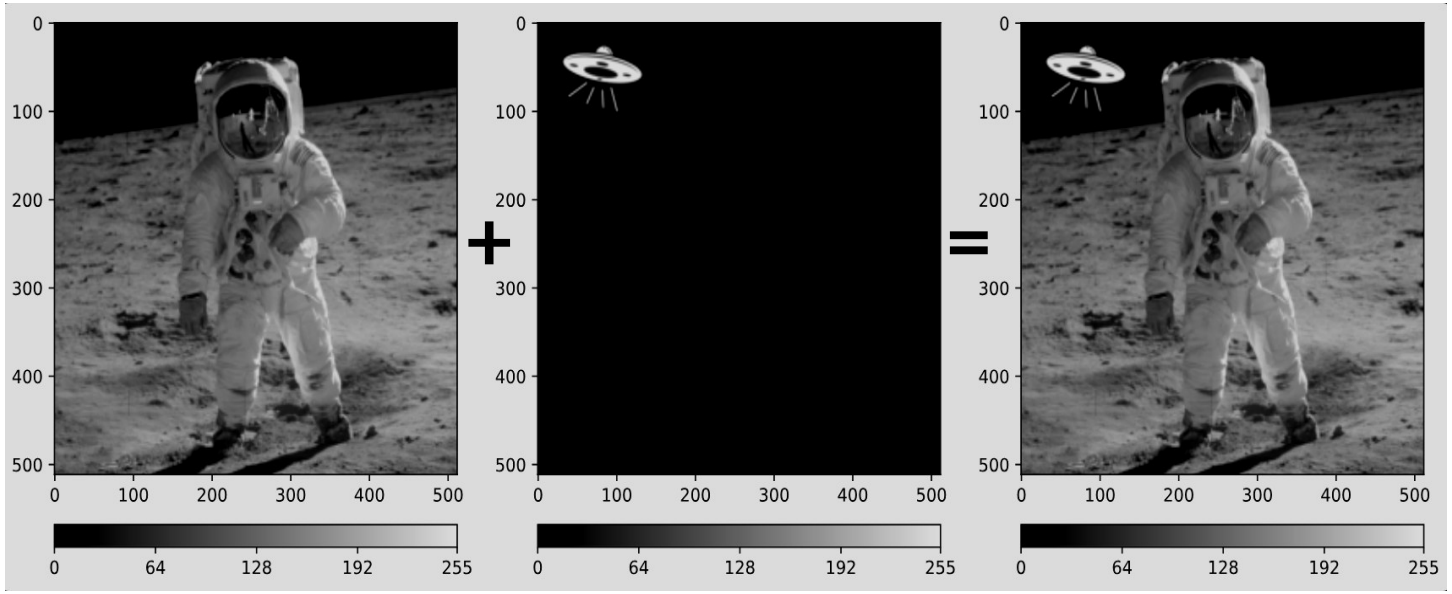


MATHEMATICAL OPERATIONS ON IMAGES

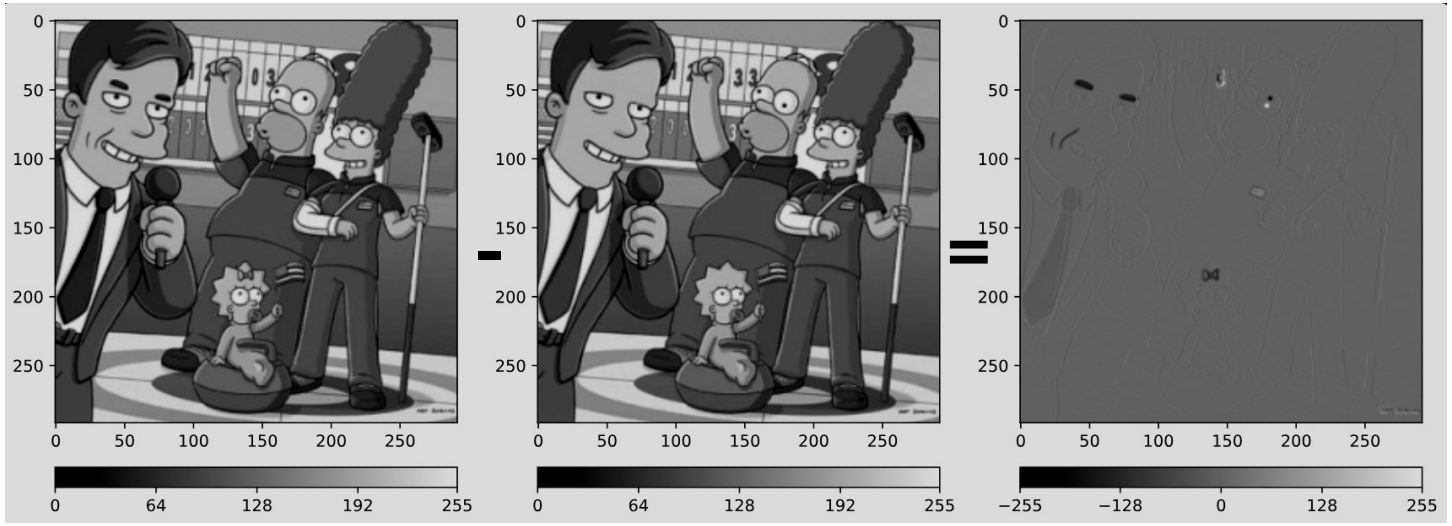
Since images can be treated as numerical matrices, we can perform the following mathematical operations:

- **Blending (Addition)**: Combines pixel intensities from multiple images to create a single composite image
- **Differencing (Subtraction)**: Computes pixel-wise differences to highlight changes between images
- **Masking and Superimposing (Bitwise Operations):** Uses binary masks and logical operations such as AND, OR, and NOT to selectively include or exclude image regions
- **Brightening and Darkening (Multiplication and Division):** Scales pixel intensity values to uniformly adjust image brightness

MATHEMATICAL OPERATIONS ON IMAGES



Addition



Subtraction

Source: Basics of image processing - Vincent Mazet

IMAGE TRANSFORMATION

When we transform an image from its original form to different forms, we apply certain mathematical operations along with a **filter** image that is also represented as a numerical matrix



FILTERS

For example, to create a sharpened version of the original image,
we add another image matrix to get the resultant image



+ a



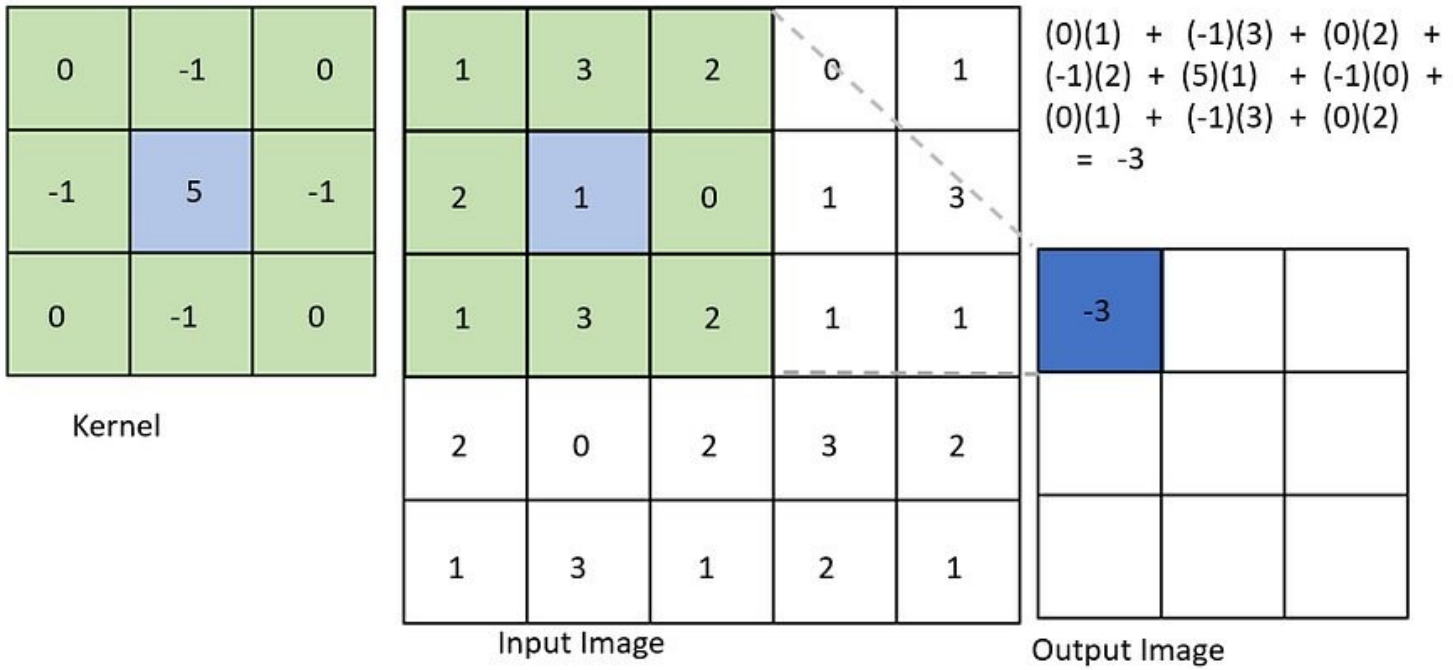
=



$$\begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 1 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} + \begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 1 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \end{bmatrix} = \begin{bmatrix} \bullet 0 & \bullet 0 & \bullet 0 \\ \bullet 0 & \bullet 2 & \bullet 0 \\ \bullet 0 & \bullet 0 & \bullet 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \\ \bullet 1 & \bullet 1 & \bullet 1 \end{bmatrix}$$

CONVOLUTION OPERATION

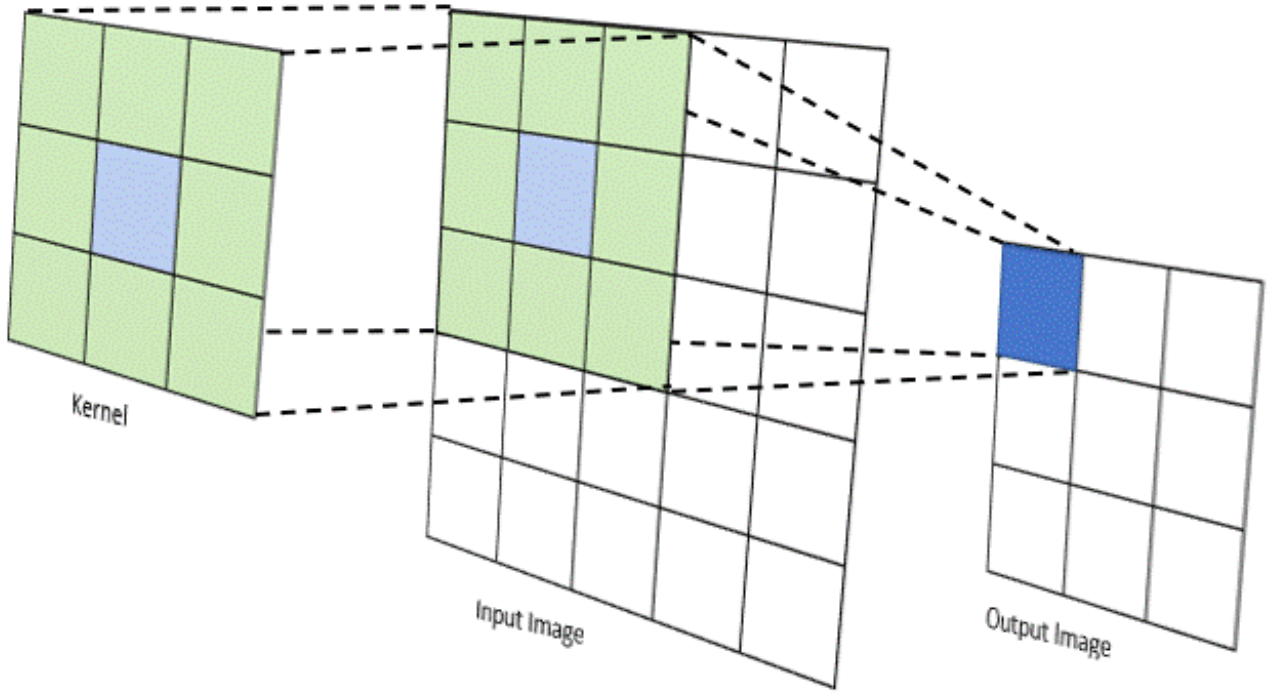
Convolution is a mathematical operation that applies a small matrix, called a **kernel** or **filter**, to an original image to transform it



It works by sliding the kernel over the image and, at each position, performing **an element-wise multiplication** with the overlapping pixels and **summing** the results to produce a new pixel value in an output image

IMAGE CONVOLUTION

The main objective of using convolution in image processing is to extract important features from the image and discard the rest. This results in a condensed representation of an image.

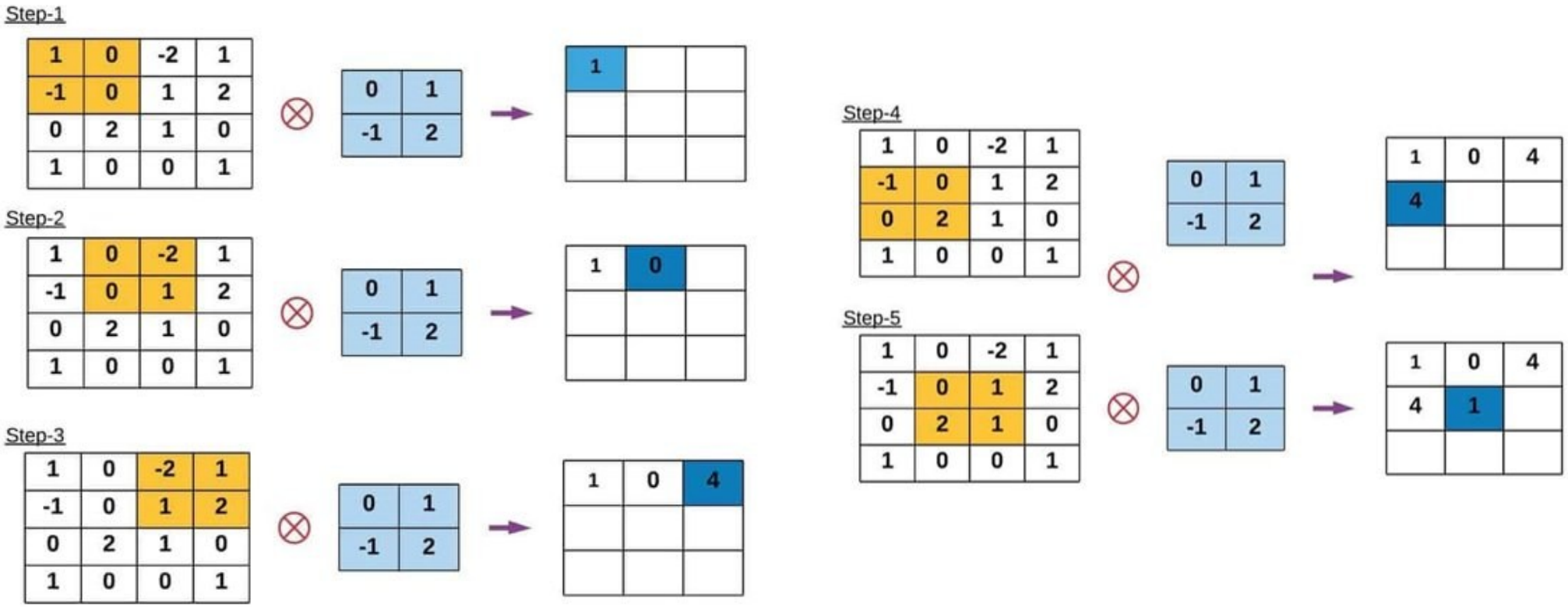


HOW CONVOLUTION WORKS

The 2D convolution of an image and a kernel computes each output pixel as the weighted sum of neighbouring image pixels defined by the kernel as it slides across the image

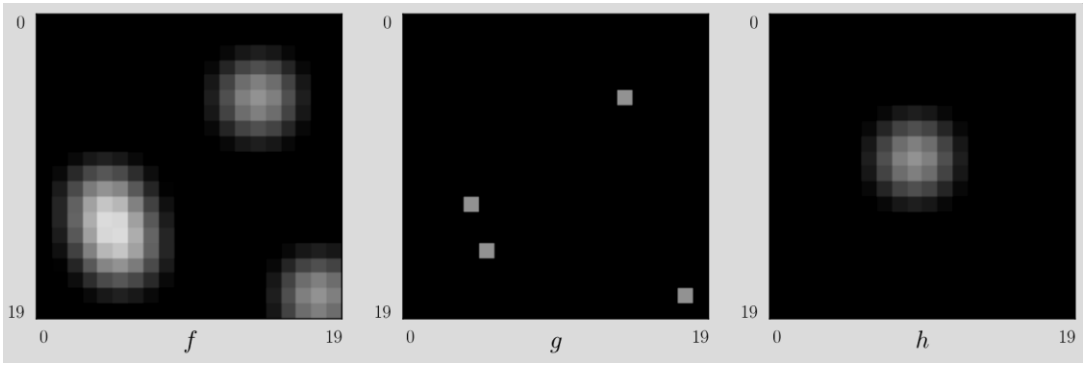
$$(I * K)(x, y) = \sum_i \sum_j I(x + i, y + j) \cdot K(i, j)$$

where (x, y) are the coordinates of the output pixel

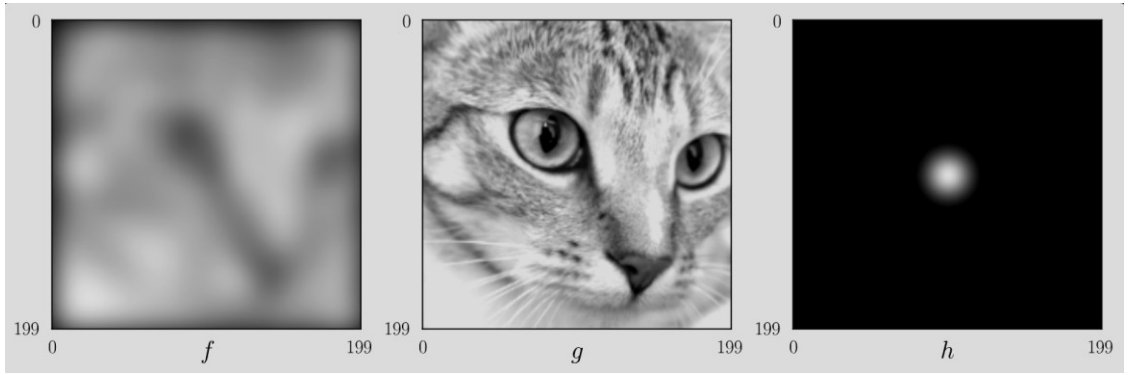


EXAMPLES

The images (f) to the left are the convolutions of the other two images (g and h)







Source: Basics of image processing - Vincent Mazet

IMAGE FILTERING METHODS

An image filter or a kernel is a matrix of a definite structure and when applying certain mathematical operators along with these filters, it can yield certain meaningful output images which is important for image analysis

ORIGINAL IMAGE



GAUSSIAN BLUR FILTER



CANNY EDGE FILTER



SOBEL FILTER



MEDIAN FILTER



GAUSSIAN BLUR FILTER

Gaussian blur uses a kernel that represents the shape of a Gaussian (bell) curve. The center pixel has the highest weight, and weights decrease as you move away from the center. This creates a smooth blurring effect.





Example 3x3 Gaussian Kernel (Approximation):

$$\frac{1}{16}\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{bmatrix}$$

Note: The 1/16 factor is a normalisation term. It ensures that the sum of all elements in the kernel is 1, which prevents the image from becoming brighter or darker during the filtering process.

SOBEL FILTER

The Sobel operator actually uses two distinct kernels: one to detect **horizontal edges** (G_x) and one to detect **vertical edges** (G_y). These are combined to find the overall edge magnitude

ORIGINAL IMAGE



$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Vertical Kernel (G_y)

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Horizontal Kernel (G_x)

SOBEL FILTER



Resulting Pixel Value: To get the final Sobel image, you calculate the magnitude (G) for each pixel:

$$\mathrm{G} = \sqrt{G_x^2 + G_y^2}$$

CANNY EDGE FILTER

Canny edge filter is not a single matrix kernel but a **multi-stage algorithm**.
We cannot define it by a single convolution matrix.

ORIGINAL IMAGE



CANNY EDGE FILTER



It involves multiple steps, such as:

- **Gaussian Blur:** To reduce noise
- **Sobel Filtering:** To find intensity gradients
- **Non-Maximum Suppression:** To thin out the edges to 1-pixel width
- **Hysteresis Thresholding:** To classify edges as "weak" or "strong" based on two threshold values

MEDIAN FILTER

The Median Filter is a **non-linear filter**, meaning it does not use a convolution matrix (kernel) and the output value is determined by sorting

ORIGINAL IMAGE



MEDIAN FILTER



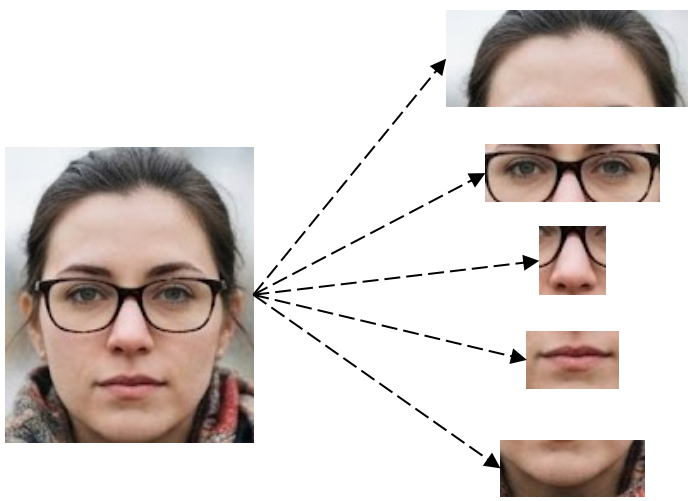
How it works:

- Select a window (e.g., 3x3) around a target pixel
- List all pixel values in that window
- Sort the values numerically
- Replace the centre pixel with the median value from the list

It is suited for removing "salt and pepper" noise while preserving edges

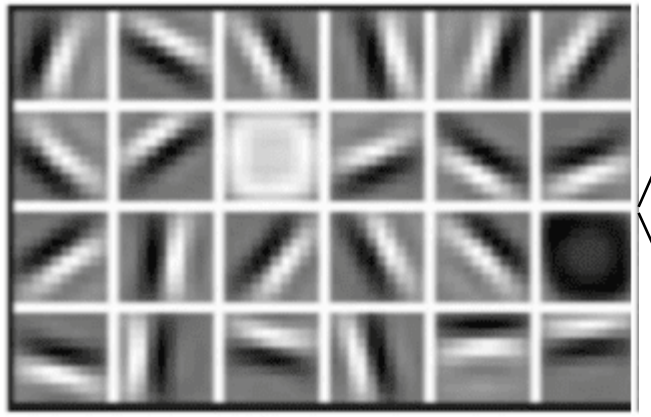
FEATURE EXTRACTION FROM IMAGES

Feature extraction is a fundamental stage in image processing and computer vision, involving the detection and representation of distinguishing structures within an image.



Blobs or
Regions of Interest (RoI)

High Level Features



Low Level Features